# Cell Painting tells you about mechanism —
# not about toxicity

**...and it beats chemical structure, which is free, at doing so.**

WS4A · cross-modal integration · HepG2

119 compounds · morphology (636) × gene expression (41,780) × chemical structure (1,024)

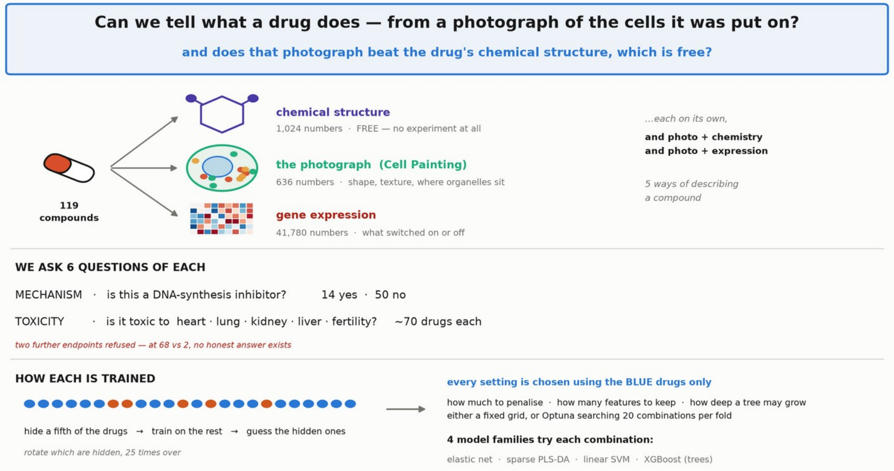
SAY THIS FIRST, OR NOTHING ELSE MAKES SENSE:

Every number in this section is an HONEST GAP — how well the model did on the real answers, minus how well the same model did when we scrambled the answers between compounds. Scrambled means there is nothing left to learn, so that second number is what luck alone buys.  Across our grid, luck scored between 0.38 and 0.60 — where 0.50 is a coin flip. That is why we never quote a raw score.

## WHAT WAS TRAINED, AND HOW

# Only 119 drugs — so every score gets a scrambled-label twin

Six questions · five kinds of data · four model families · every setting chosen on drugs the score never sees.




**THE DATA.** 119 compounds, each described three ways: its chemical structure (free — computed from the molecule, no experiment); the Cell Painting photograph of treated HepG2 cells (636 measurements of shape, texture and where organelles sit); and gene expression (41,780 genes). The expression comes from 384,533 single cells, but cells given the same drug are replicates — they are averaged into ONE row per drug. So the model sees 119 examples, not 384,000.

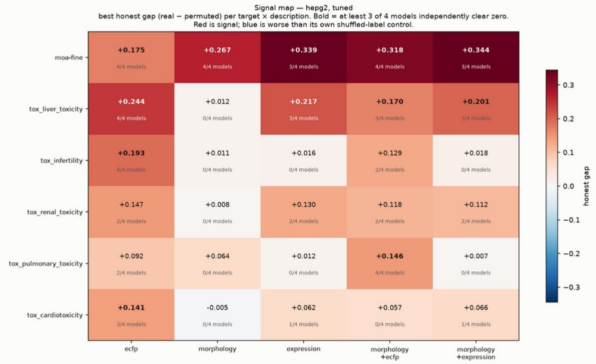
**WHAT WE PREDICT.** Six questions: is this a DNA-synthesis inhibitor (14 yes vs 50 no), and is it toxic to heart, lung, kidney, liver or fertility (~70 drugs each). Two further toxicity endpoints were refused outright — at 68 yes vs 2 no, a model that always says yes is right 97% of the time and has learned nothing.

**HOW A MODEL IS TESTED.** Hide a fifth of the drugs, learn from the rest, guess the hidden ones, and rotate which are hidden 25 times. Every setting — how much to penalise, how many features to keep, how deep a tree may grow — is chosen using only the training drugs, so the ones being scored never influence anything.

**WHY EVERYTHING RUNS TWICE.** With 119 rows and up to 41,780 measurements, some measurements line up with the answer by pure chance. So we repeat the whole thing with the answers SCRAMBLED between drugs — nothing left to learn. Luck alone scored 0.38 to 0.60, where 0.50 is a coin flip. Every number on the next slides is real minus scrambled.

# Morphology beats chemistry — for mechanism only

+0.143 over chemical structure for mechanism of action, all four models agreeing.  For all five toxicity endpoints it adds nothing.




**WHAT A "MODEL" IS.** A recipe for guessing. We show it drugs whose answer we already know — here are 636 numbers from the photograph, and yes, this one is a DNA-synthesis inhibitor — and it learns a rule. Then it must guess drugs it has NEVER seen. Four different recipes are used on every box: three that draw a dividing line in different ways (elastic net, sparse PLS-DA, linear SVM) and one that builds a flowchart of yes/no questions instead (XGBoost).

**SHUFFLED vs UNSHUFFLED — every box is run TWICE.** UNSHUFFLED: each drug keeps its true answer. SHUFFLED: we deal the answers out at random, so drug A is given drug B's answer — the link between measurements and answer is destroyed on purpose, so there is nothing left to learn. The model still scores something, because with 119 drugs and up to 41,780 measurements some line up with random answers by pure chance. THAT is the luck level, and it differs for every box.

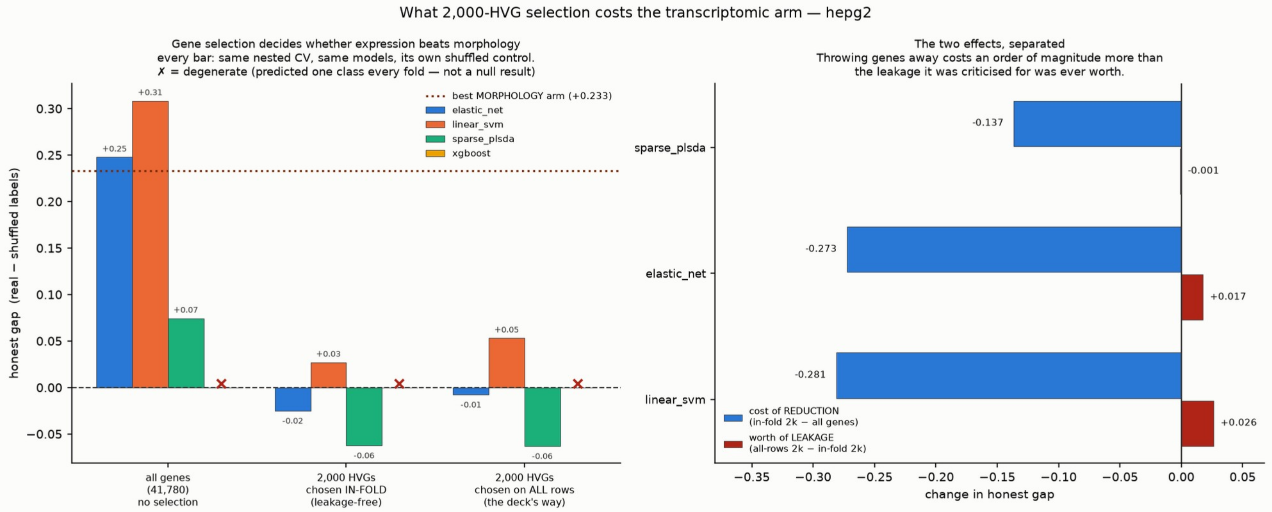
**WHERE THE NUMBER IN EACH BOX COMES FROM.** Worked example — morphology predicting mechanism, the box reading +0.267. The best model got 74.5 % right on the real answers. With the answers shuffled it still got 47.7 %. 0.745 − 0.477 = +0.267, and that difference is what is printed. So the number is NOT accuracy — it is how far the model beat its own shuffled twin. Note the shuffled run was 47.7 %, not 50 %: across the four models in that one box it ranged 47.7–51.7 %, which is why each box is compared against its OWN control rather than against 0.5.

**HOW TO READ THE GRID.** Rows are the six questions; columns are what the model was allowed to look at. Red = the model beat its shuffled twin. White = it did not. −0.005 means it did slightly WORSE than nonsense, which is what "nothing here" looks like when measured honestly. The small print — 4/4 models — is how many of the four recipes each independently beat its own shuffled run.

**WHAT IT SHOWS, AND THE POINT.** The top row, mechanism of action, is the only row that works across the board: chemistry +0.175, the photograph +0.267, both together +0.318. Read DOWN the morphology column and every toxicity row is near-white with 0 of 4 models finding anything — while the chemistry column IS predicting them. Chemistry is the control because it is free, so the question is not whether imaging beats chance but whether it beats the free option. For MECHANISM it does, by +0.143. For TOXICITY it does not.

# Gene selection, not the modality, is why expression lost

Dropping to 2,000 highly variable genes costs 16× what the leakage it is usually criticised for was worth.



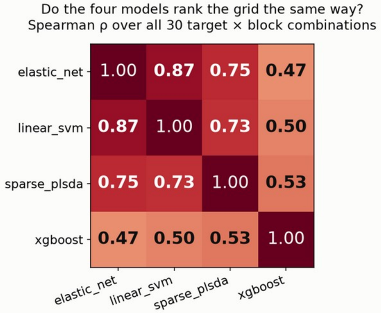

**THE BACKGROUND.** Almost every single-cell workflow keeps only the ~2,000 most variable genes out of 40,000. Task 1 of this workstream does that, and its own caveat slide flags the leakage. But that caveat bundles TWO different problems: REDUCTION (throwing 39,780 genes away) and LEAKAGE (choosing which 2,000 using drugs you later test on). Nobody knew which was actually costing anything.

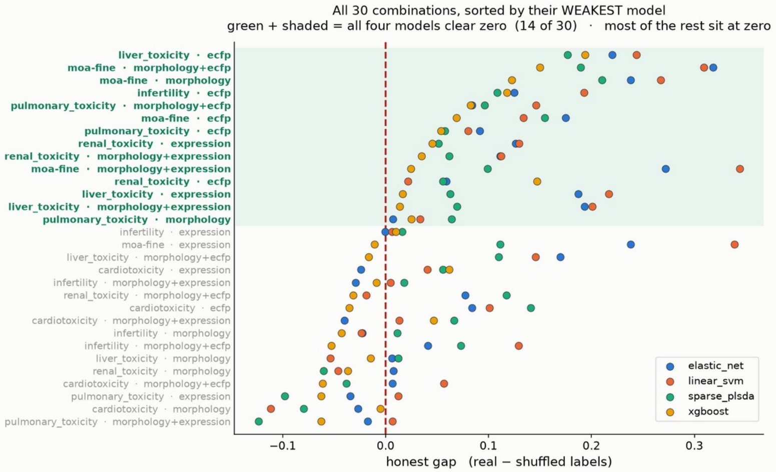
**HOW TO READ IT.** Left: three groups of bars, identical cross-validation in each — all 41,780 genes; 2,000 genes chosen separately INSIDE each training fold (leakage-free); and 2,000 chosen once using all drugs (the bundled approach). The brown dotted line is the best morphology result. Right: the two effects separated — blue is what reduction costs, red is what the leakage was worth.

**THE POINT.** Reduction costs −0.273; the leakage was worth +0.017. About sixteen times more. And with all 41,780 genes the expression arm clears the brown line — with 2,000 it does not come close. So "gene expression does not beat morphology" looks like a property of the gene selection, not of the modality. Both pipelines agree once the same genes are used.

# Four different models, the same answer

Model families that fail in different ways agree on where the signal is — better evidence here than a p-value cross-validation cannot honestly give.




**WHY THIS SLIDE.** The obvious objection to any result from 119 compounds is that we got lucky with one model. Our uncertainty intervals are also anti-conservative — cross-validation folds share training data, so there is no honest variance estimate at this sample size. So instead of a fragile p-value we show four model families that fail in different ways, and ask whether they agree.

**HOW TO READ IT.** Left: how similarly each pair of models ranks all 30 question × data combinations (1.00 = identical ordering, 0 = unrelated). Right: every combination as one row with all four models plotted, sorted by its WEAKEST model. The red line is zero — no better than that model's own scrambled control. Green and shaded = all four models clear zero.

**THE POINT.** The three linear models agree closely (0.73–0.87). XGBoost, which builds decision trees rather than drawing a line, agrees far less (0.47–0.53) — and that is exactly why its agreement counts: it is looking for a different kind of pattern. In 14 of the 30 combinations all four models clear zero, and that green set includes mechanism from morphology. Most of the other 16 sit at zero: the signal is concentrated, not everywhere.

# Narrow claims, and the controls that made them narrow

Several of these controls changed our own conclusions — that is the point.

WE CAN SAY

- Morphology adds +0.143 over chemical structure for mechanism, 4 of 4 models
- It adds nothing for the five toxicity endpoints — chemistry does that
- Fusion does not help — we independently replicate the main analysis
- The modalities agree weakly but really (adjusted RV 0.016, Mantel p = 0.0001)
- Dropping to 2,000 HVGs costs 16× what its leakage was worth

WE MUST NOT SAY

- Any plain RV number — it scores random noise as high as our real data
- "r = 0.903" — its own null reaches 0.902. Quote the p-value (0.017)
- "Four joint components" — the null also reached four, p = 0.095
- Any gene list — 0 of 41,780 survive
- Anything about A549 — its morphology is contaminated (max 1.5e19)

**Every number is shown next to the same number computed on data where the answer is known to be nothing —**
**and several of those controls changed our conclusions.**